Weizheng Wang

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Research profile

Applied AI researcher working on multimodal model evaluation, efficient visual perception and edge deployment. I build reproducible experimental pipelines, develop controlled benchmarks and deploy models on edge hardware, with applications in UAV inspection and industrial monitoring.

Education

Adelaide University 2025–2026
Master of Artificial Intelligence and Machine Learning | Conferred 7 October 2026

GPA: **6.75/7.0**; Research Project A: **90**, B: **87** (High Distinction). Completed the two-year programme within one year. Global Citizens International Scholarship (highest tier).

North China Institute of Science and Technology 2021–2025
BEng in Computer Science and Technology | GPA: **4.02/5.0**

Research and engineering experience

Zayed University Feb.–Aug. 2026
Research Assistant | Remote

- **MulRobBench:** co-designed a decision-level benchmark for safe, security-policy-compliant multimodal UAV agents. Built data-generation, inference and evaluation pipelines covering **3,024 samples and 17 models**; analysed semantic decisions, structural compliance and unsafe actions.
- Implemented controlled visual degradation and noisy-instruction conditions to examine failure modes under matched inputs. Contributed methodology, software, formal analysis and manuscript writing.

Institute of Software, Chinese Academy of Sciences Feb.–Jun. 2025
Research Intern | MindNLP / MindSpore

Implemented and fine-tuned **Autoformer, BEiT and ALBERT**, spanning time-series, vision and language models. Worked on training stability, reproducible evaluation, code reviews and integration documentation.

Beijing Elite Intelligence Company and NCIST Jan.–Jul. 2024
UAV-based mine inspection and industrial vision

Implemented standing-water counting, belt-deviation detection and truck monitoring from UAV video using the **YOLO series and DeepSORT**. Optimised PyTorch detection pipelines and ONNX export for downstream deployment.

Selected technical projects

DEIM Jetson acceleration May 2026–present
Built ONNX/TensorRT conversion, output-correctness checks and device-level latency measurement on Jetson. Diagnosed memory-layout and quantisation issues; in a GPU-resident batch-one 640-by-640 engine trial, measured **89.48 FPS (FP16)** versus **82.59 FPS (selective INT8)** after 50 warm-ups and 500 synchronised iterations.

Jetson VLM Lab
Built configuration-driven model launch and a shared Python client for text, image and single-frame VLM tests. Compared GPU offload, batch and projector placement; recorded timing, model settings and errors in JSONL manifests.

Master's research: graph contraction and visual exploration
Built an occupation–industry graph and Louvain contraction pipeline, reducing nodes from **1,255 to 609**; developed interactive search, source-evidence views and graph-grounded explanations, and evaluated graph properties and technical behaviour.

Substation alarm detection system (BEng thesis)
Integrated multistream video, open-vocabulary detection, alarm handling and vision–language question answering into a working inspection system.

Publications

A Multi-Source Data Fusion and PSO-Transformer Decision Support System for Gold Price Volatility Forecasting and Risk Warning. *Mathematics*, 2026, 14(21), 4115. Co-author; code implementation. DOI: 10.3390/math14214115.

A Lightweight Thermal Denoising and Occlusion-Robust Infrared Detection Model for Substation Equipment. IEEE ICIP, 2026. **Co-first author.**
Responsible for algorithm improvements, the model-training pipeline, experiments and writing the experimental section.

A lightweight insulator defect detection algorithm based on drone images for power line inspection. *Engineering Research Express*, 2025. Co-author. DOI: 10.1088/2631-8695/ae19ce.

Real-time monitoring of trucks used in open pit based on aerial video of UAV. IEEE RAIIC, 2024, pp. 130–138. **First author**; code implementation and manuscript writing.

Research on industrial meter readings based on improved YOLOv8. SPIE, 2024, 13181, 131815G. Co-author; code implementation.

Preprints and submitted manuscripts

MulRobBench: A Decision-Level Benchmark for Safe and Security-Policy-Compliant Multimodal UAV Agents. Co-first author, with Belal S. Alsinglawi, Junyi Wu, Lianhai Lin, Mérouane Debbah and Izzat Alsmadi. arXiv preprint, 2026. arXiv:2607.23870.

FHS-DEIM: Frequency-Aware Hybrid Encoding with Spiral Mamba for Infrared UAV Search and Rescue. Co-first author, with Junyi Wu, Yi Jiang, Yong Niu, Liqin Tian and Belal S. Alsinglawi. Submitted to *Journal of King Saud University Computer and Information Sciences*, 2026. Code implementation and manuscript writing.

H2T-DEIM: Hilbert–Haar Representation with Tri-Modal Multi-Dimensional Pruning for Maritime Small-Object Detection from Uncrewed Aerial Vehicle Imagery. Co-first author. Manuscript, 2026. Methodology, software implementation and validation.

TRIM: Dual-Budget Structured Pruning with Information-Modulated Saliency for CNN, Transformer, and Mamba Vision Models. Co-first author. Submitted to IEEE ICASSP 2027. Implemented dependency-aware pruning and dual-budget optimisation; contributed manuscript writing.

When Retrieval Is Not Binding: A Diagnostic Benchmark for Visual Episodic Memory. Co-first author. Submitted to IEEE ICASSP 2027. Implemented controlled memory-off, wrong-entity and shuffled-binding comparisons; contributed manuscript writing.

EmergUAV-Bench: Protocol-Conditioned Evaluation of Vision-Language-Action Models for Emergency UAV Decisions. Co-first author. Submitted to IEEE ICASSP 2027. Code implementation and writing; 1,280 samples, 13 models and six decision tasks.

ManiSeg: A Semantic-Entropy Driven Manifold Collision Framework for High-Dimensional Feature Evaluation of Detectors. Co-first author. Submitted to IEEE ICASSP 2027. Code implementation for feature evaluation and semantic-entropy-based analysis.

Technical skills

Python, C/C++; PyTorch, MindSpore/MindNLP; multimodal and VLA evaluation; structured pruning; YOLO-series detection and tracking; ONNX, TensorRT and Jetson deployment; reproducible experimental pipelines, Git and Linux.